

Nwaogugu Chibuike Collins

Senior AI Engineer · Remote

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 8+ years of progressive software engineering experience, specialising in production multi-agent systems, MLOps, and AI developer tooling. Proven track record architecting and deploying large-scale AI platforms on AWS — from autonomous research agents and semantic knowledge bases to org-wide observability stacks. Thrives in async-first, globally distributed teams; known for translating complex AI infrastructure into measurable business impact.

CORE COMPETENCIES

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|-------------------------------------|--|--|
| • Multi-Agent Systems (AWS Bedrock) | • RAG Pipelines & Semantic Search | • MLOps & Model Observability |
| • AWS (Bedrock, SageMaker, Lambda) | • LangChain / LangGraph / OpenAI SDK | • Terraform / IaC / API-first Design |
| • .NET Core / Python / ASP.NET | • Event-Driven Microservices (SQS/SNS) | • OpenTelemetry / Prometheus / Grafana |

PROFESSIONAL EXPERIENCE

Senior AI Engineer · Confidential / Contract

Feb 2026 – May 2026

AI Developer Tooling · Multi-Agent Platforms · MLOps · AWS

- Designed and deployed org-wide AI developer tooling and evaluation harnesses used across engineering, product, marketing, and operations — integrating and configuring third-party AI tools (coding assistants, LLM orchestration frameworks, multi-agent platforms) to match evolving organisational needs; raised the bar for how engineering teams used AI coding agents in daily workflows.
- CloudWatch · OpenTelemetry · Prometheus/Grafana:**Built org-wide AI usage dashboard and telemetry layer using
- Tracking adoption metrics, inference latency, token consumption, and tooling health across Bedrock, OpenAI, and SageMaker endpoints — giving leadership real-time visibility into AI productivity and model performance.
- Alex (Multi-Agent SaaS Platform):**Architected and deployed Alex, a production multi-agent SaaS platform on AWS using API-first design:
- 7 Terraform modules provisioning S3 Vectors, Aurora Serverless v2, SageMaker embedding endpoints, Lambda agent functions (Planner, Tagger, Reporter, Charter, Retirement), SQS job queues, API Gateway, CloudFront CDN, and App Runner containerised services.
- AI Research Agent:**Built production AI research agent using OpenAI Agents SDK, AWS Bedrock Nova Pro, and Playwright MCP for autonomous web browsing and real-time data ingestion; developed entirely with Claude Code and Cursor — reducing feature delivery time by ~40% while maintaining full test coverage; results stored in S3 Vectors semantic knowledge base via EventBridge scheduled runs every 2 hours.
- No-Code ML Pipelines:**Built configuration-driven ML data pipelines (JSON-schema orchestration via Apache Airflow + AWS SQS/EventBridge) enabling non-engineers to modify ingestion and processing rules without code deploys — reducing manual engineering overhead by 60%+ and cutting pipeline onboarding from days to hours.
- MLOps:**Implemented MLOps best practices: automated model retraining triggers with drift detection, CI/CD-integrated Lambda deployments, and inference latency / token usage monitoring across Bedrock, OpenAI, and SageMaker endpoints.

- **NLP & ML Features:** Developed NLP-powered features including semantic search, RAG pipelines, structured output parsing, and prompt engineering using LangChain, LangGraph, and OpenAI API; applied TensorFlow and Scikit-learn for transaction anomaly detection.
- **Internal Knowledge Assistant:** Built an internal knowledge assistant over markdown docs (chunking → embedding → Chroma retrieval → conversational Q&A) with an evaluation harness tracking retrieval metrics (MRR/nDCG) and answer quality scoring.
- **Async-First Execution:** Operated as a fully self-directed engineer in an async-first, globally distributed team — proactively communicating project status, flagging blockers early, and aligning cross-functional stakeholders through structured async documentation and architecture design-review rituals.

Senior Software Engineer · Deimos

Apr 2022 – Feb 2026

Microservices · AWS · Observability · CI/CD · Kubernetes

- Designed and implemented event-driven microservices using ASP.NET Core and AWS EventBridge/SQS, applying Outbox patterns for reliable messaging at scale.
- Established comprehensive testing strategies — unit, integration, and end-to-end — using xUnit, Testcontainers, and LocalStack, significantly improving release confidence and regression detection.
- Managed deployment pipelines with GitLab CI/Jenkins, Docker, and Helm for Kubernetes, improving release reliability, cadence, and rollback capability.
- Enhanced system observability using OpenTelemetry, Prometheus/Grafana, and Sentry for real-time monitoring and alerting across distributed services.
- Spearheaded integration of cloud technologies, optimising resource allocation and infrastructure management across production environments.

Software Engineer · Andela

Jan 2021 – Dec 2022

Backend Engineering · CI/CD · AI/ML Evaluation · Mentorship

- Developed high-performance backend features using .NET Core and Python across multiple product squads, focusing on scalability, data-processing throughput, and consumer-facing API reliability.
- Built and optimised CI/CD pipelines improving deployment frequency and reducing regression rates; introduced automated testing frameworks that significantly cut release cycle time.
- Evaluated emerging AI/ML technologies to inform strategic architecture decisions and mentored team members through structured code reviews and pair programming in a fully async-first remote environment.

Senior Software Engineer · Venture Garden Group

Jan 2017 – Dec 2021

Microservices · Legacy Modernisation · PCI/DSS · CI/CD

- Built and deployed scalable microservices and legacy-modernisation projects using ASP.NET Core and Python, laying data infrastructure groundwork later leveraged for ML and AI workloads.
- Led integration of Central Reservation System (CRS) ensuring full PCI/DSS compliance across payment processing flows.
- Implemented CI/CD processes to standardise and accelerate deployments across distributed, multi-team engineering organisations.

TECHNOLOGY STACK

AI / ML	OpenAI API, AWS Bedrock (Nova Pro), SageMaker, LangChain, LangGraph, OpenAI Agents SDK, Playwright MCP, TensorFlow, Scikit-learn, Chroma, S3 Vectors
Cloud & Infra	AWS (Lambda, SQS, EventBridge, API Gateway, CloudFront, App Runner, Aurora Serverless v2, S3), Terraform, Docker, Kubernetes, Helm
Observability	OpenTelemetry, Prometheus, Grafana, CloudWatch, Sentry
Data & Pipelines	Apache Airflow, AWS SQS/EventBridge, JSON-schema orchestration
Backend	.NET Core / ASP.NET Core, Python, microservices, event-driven architecture, Outbox pattern
DevOps / CI/CD	GitLab CI, Jenkins, GitHub Actions, Claude Code, Cursor

Testing

xUnit, Testcontainers, LocalStack, MRR/nDCG evaluation harnesses